



Data Output from F1® 25: 2026 Season Pack

Version 1.2

Added F2 Driver IDs as part of the Season 8 "Sports Update".

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Overview

The F1® 25: 2026 Season Pack supports the output of certain game data across UDP connections. This data can be used supply race information to external applications, or to drive certain hardware (e.g. motion platforms, force feedback steering wheels and LED devices).

The following information summarises these data structures so that developers of supporting hardware or software can configure these to work correctly with the F1® 25: 2026 Season Pack.

Note: To ensure that you are using the latest specification for this game, please check our official forum page [here](#). For questions on the specification, please use the thread [here](#). For suggestions for future titles please use the thread [here](#).

If you cannot find the information that you require then please contact the team via the official forum thread listed above. For any bugs with the UDP system, please post a new bug report on the F1® 25 Game forum in the appropriate threads above.

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Packet Information

Packet Types

Each packet carries different types of data rather than having one packet which contains everything. The header in each packet describes the packet type and versioning info so it will be easier for applications to check they are interpreting the incoming data in the correct way. Please note that all values are encoded using Little Endian format. All data is packed.

The following data types are used in the structures:

Type	Description
uint8	Unsigned 8-bit integer
int8	Signed 8-bit integer
uint16	Unsigned 16-bit integer
int16	Signed 16-bit integer
uint32	Unsigned 32-bit integer
float	Floating point (32-bit)
double	Double-precision floating point (64-bit)
uint64	Unsigned 64-bit integer
char	Character

Packet Header

Each packet has the following header:

```
struct PacketHeader
{
    uint16    m_packetFormat;           // 2026
    uint8     m_gameYear;               // Game year - last two digits e.g. 25
    uint8     m_gameMajorVersion;       // Game major version - "X.00"
    uint8     m_gameMinorVersion;       // Game minor version - "1.XX"
    uint8     m_packetVersion;          // Version of this packet type, all start from 1
    uint8     m_packetId;               // Identifier for the packet type, see below
    uint64    m_sessionUID;             // Unique identifier for the session
    float     m_sessionTime;            // Session timestamp
    uint32    m_frameIdentifier;         // Identifier for the frame the data was retrieved on
    uint32    m_overallFrameIdentifier; // Overall identifier for the frame the data was retrieved
                                         // on, doesn't go back after flashbacks
    uint8     m_playerCarIndex;         // Index of player's car in the array
    uint8     m_secondaryPlayerCarIndex; // Index of secondary player's car in the array (splitscreen)
                                         // 255 if no second player
};
```

Packet IDs

The packets IDs are as follows:

Packet Name	Value	Description
Motion	0	Contains all motion data for player's car – only sent while player is in control
Session	1	Data about the session – track, time left
Lap Data	2	Data about all the lap times of cars in the session
Event	3	Various notable events that happen during a session



Participants	4	List of participants in the session, mostly relevant for multiplayer
Car Setups	5	Packet detailing car setups for cars in the race
Car Telemetry	6	Telemetry data for all cars
Car Status	7	Status data for all cars
Final Classification	8	Final classification confirmation at the end of a race
Lobby Info	9	Information about players in a multiplayer lobby
Car Damage	10	Damage status for all cars
Session History	11	Lap and tyre data for session
Tyre Sets	12	Extended tyre set data
Motion Ex	13	Extended motion data for player car
Time Trial	14	Time Trial specific data
Lap Positions	15	Lap positions on each lap so a chart can be constructed
Car Telemetry 2	16	Additional telemetry data for all cars

Motion Packet

The motion packet gives physics data for all the cars being driven.

N.B. For the normalised vectors below, to convert to float values divide by 32767.0f – 16-bit signed values are used to pack the data and on the assumption that direction values are always between -1.0f and 1.0f.

Also, for the g-Force components, these are int16 and need to be converted by dividing by 1000.0f.

Frequency: Rate as specified in menus

Size: 1325 bytes

Version: 1

```
struct CarMotionData
{
    float        m_worldPositionX;           // World space X position - metres
    float        m_worldPositionY;           // World space Y position
    float        m_worldPositionZ;           // World space Z position
    float        m_worldVelocityX;           // Velocity in world space X - metres/s
    float        m_worldVelocityY;           // Velocity in world space Y
    float        m_worldVelocityZ;           // Velocity in world space Z
    int16        m_worldForwardDirX;          // World space forward X direction (normalised)
    int16        m_worldForwardDirY;          // World space forward Y direction (normalised)
    int16        m_worldForwardDirZ;          // World space forward Z direction (normalised)
    int16        m_worldRightDirX;           // World space right X direction (normalised)
    int16        m_worldRightDirY;           // World space right Y direction (normalised)
    int16        m_worldRightDirZ;           // World space right Z direction (normalised)
    int16        m_gForceLateral;            // Lateral G-Force component (quantised)
    int16        m_gForceLongitudinal;       // Longitudinal G-Force component (quantised)
    int16        m_gForceVertical;          // Vertical G-Force component (quantised)
    float        m_yaw;                     // Yaw angle in radians
    float        m_pitch;                   // Pitch angle in radians
    float        m_roll;                    // Roll angle in radians
};

struct PacketMotionData
{
    PacketHeader  m_header;                 // Header

    CarMotionData m_carMotionData[24];      // Data for all cars on track
};
```



Session Packet

The session packet includes details about the current session in progress.

Frequency: 2 per second

Size: 926 bytes

Version: 1

```
struct MarshalZone
{
    float m_zoneStart;    // Fraction (0..1) of way through the lap the marshal zone starts
    int8 m_zoneFlag;      // -1 = invalid/unknown, 0 = none, 1 = green, 2 = blue, 3 = yellow
};

struct ActiveAeroZone
{
    float m_zoneStart;    // Fraction (0..1) of way through the lap the Active Aero zone starts
    float m_zoneEnd;      // Fraction (0..1) of way through the lap the Active Aero zone ends
};

struct DRSZone
{
    float m_zoneStart;    // Fraction (0..1) of way through the lap the DRS zone starts
    float m_zoneEnd;      // Fraction (0..1) of way through the lap the DRS zone ends
};

struct WeatherForecastSample
{
    uint8 m_sessionType;  // 0 = unknown, see appendix
    uint8 m_timeOffset;   // Time in minutes the forecast is for
    uint8 m_weather;      // Weather - 0 = clear, 1 = light cloud, 2 = overcast
                        // 3 = light rain, 4 = heavy rain, 5 = storm
    int8 m_trackTemperature; // Track temp. in degrees Celsius
    int8 m_trackTemperatureChange; // Track temp. change - 0 = up, 1 = down, 2 = no change
    int8 m_airTemperature;  // Air temp. in degrees celsius
    int8 m_airTemperatureChange; // Air temp. change - 0 = up, 1 = down, 2 = no change
    uint8 m_rainPercentage; // Percentage chance of rain (0-100)
};

struct PacketSessionData
{
    PacketHeader m_header; // Header

    uint8 m_weather; // Weather - 0 = clear, 1 = light cloud, 2 = overcast
                        // 3 = light rain, 4 = heavy rain, 5 = storm
    int8 m_trackTemperature; // Track temp. in degrees celsius
    int8 m_airTemperature;  // Air temp. in degrees celsius
    uint8 m_totalLaps;      // Total number of laps in this race
    uint16 m_trackLength;   // Track length in metres
    uint8 m_sessionType;    // 0 = unknown, see appendix
    int8 m_trackId;         // -1 for unknown, see appendix
    uint8 m_formula;        // Formula, 0 = F1 Modern, 1 = F1 Classic, 2 = F2,
                        // 3 = F1 Generic, 4 = Beta, 6 = Esports
                        // 8 = F1 World, 9 = F1 Elimination, 13 = F1 26
    uint16 m_sessionTimeLeft; // Time left in session in seconds
    uint16 m_sessionDuration; // Session duration in seconds
    uint8 m_pitSpeedLimit;   // Pit speed limit in kilometres per hour
    uint8 m_gamePaused;      // Whether the game is paused - network game only
    uint8 m_isSpectating;    // Whether the player is spectating
    uint8 m_spectatorCarIndex; // Index of the car being spectated
    uint8 m_sliProNativeSupport; // SLI Pro support, 0 = inactive, 1 = active
    uint8 m_numMarshalZones; // Number of marshal zones to follow
    MarshalZone m_marshalZones[21]; // List of marshal zones - max 21
    uint8 m_safetyCarStatus; // 0 = no safety car, 1 = full
                        // 2 = virtual, 3 = formation lap
    uint8 m_networkGame;     // 0 = offline, 1 = online
    uint8 m_numWeatherForecastSamples; // Number of weather samples to follow
    WeatherForecastSample m_weatherForecastSamples[64]; // Array of weather forecast samples
    uint8 m_forecastAccuracy; // 0 = Perfect, 1 = Approximate
    uint8 m_aiDifficulty;     // AI Difficulty rating - 0-110
};
```



```
uint32      m_seasonLinkIdentifier;    // Identifier for season - persists across saves
uint32      m_weekendLinkIdentifier;    // Identifier for weekend - persists across saves
uint32      m_sessionLinkIdentifier;    // Identifier for session - persists across saves
uint8       m_pitStopWindowIdealLap;    // Ideal lap to pit on for current strategy (player)
uint8       m_pitStopWindowLatestLap;   // Latest lap to pit on for current strategy (player)
uint8       m_pitStopRejoinPosition;    // Predicted position to rejoin at (player)
uint8       m_steeringAssist;           // 0 = off, 1 = on
uint8       m_brakingAssist;            // 0 = off, 1 = low, 2 = medium, 3 = high
uint8       m_gearboxAssist;            // 1 = manual, 2 = manual & suggested gear, 3 = auto
uint8       m_pitAssist;                // 0 = off, 1 = on
uint8       m_pitReleaseAssist;         // 0 = off, 1 = on
uint8       m_ERSAssist;                // 0 = off, 1 = on
uint8       m_DRSAssist;                // 0 = off, 1 = on
uint8       m_dynamicRacingLine;        // 0 = off, 1 = corners only, 2 = full
uint8       m_dynamicRacingLineType;    // 0 = 2D, 1 = 3D
uint8       m_gameMode;                 // Game mode id - see appendix
uint8       m_ruleSet;                  // Ruleset - see appendix
uint32      m_timeOfDay;                 // Local time of day - minutes since midnight
uint8       m_sessionLength;            // 0 = None, 2 = Very Short, 3 = Short, 4 = Medium
                                                // 5 = Medium Long, 6 = Long, 7 = Full

uint8       m_speedUnitsLeadPlayer;     // 0 = MPH, 1 = KPH
uint8       m_temperatureUnitsLeadPlayer; // 0 = Celsius, 1 = Fahrenheit
uint8       m_speedUnitsSecondaryPlayer; // 0 = MPH, 1 = KPH
uint8       m_temperatureUnitsSecondaryPlayer; // 0 = Celsius, 1 = Fahrenheit
uint8       m_numSafetyCarPeriods;       // Number of safety cars called during session
uint8       m_numVirtualSafetyCarPeriods; // Number of virtual safety cars called
uint8       m_numRedFlagPeriods;        // Number of red flags called during session
uint8       m_equalCarPerformance;      // 0 = Off, 1 = On
uint8       m_recoveryMode;              // 0 = None, 1 = Flashbacks, 2 = Auto-recovery
uint8       m_flashbackLimit;            // 0 = Low, 1 = Medium, 2 = High, 3 = Unlimited
uint8       m_surfaceType;              // 0 = Simplified, 1 = Realistic
uint8       m_lowFuelMode;              // 0 = Easy, 1 = Hard
uint8       m_raceStarts;               // 0 = Manual, 1 = Assisted
uint8       m_tyreTemperature;          // 0 = Surface only, 1 = Surface & Carcass
uint8       m_pitLaneTyreSim;           // 0 = On, 1 = Off
uint8       m_carDamage;                // 0 = Off, 1 = Reduced, 2 = Standard, 3 = Simulation
uint8       m_carDamageRate;            // 0 = Reduced, 1 = Standard, 2 = Simulation
uint8       m_collisions;               // 0 = Off, 1 = Player-to-Player Off, 2 = On
uint8       m_collisionsOffForFirstLapOnly; // 0 = Disabled, 1 = Enabled
uint8       m_mpUnsafePitRelease;       // 0 = On, 1 = Off (Multiplayer)
uint8       m_mpOffForGriefing;         // 0 = Disabled, 1 = Enabled (Multiplayer)
uint8       m_cornerCuttingStringency;   // 0 = Regular, 1 = Strict
uint8       m_parcFerreRules;           // 0 = Off, 1 = On
uint8       m_pitStopExperience;        // 0 = Automatic, 1 = Broadcast, 2 = Immersive
uint8       m_safetyCar;                // 0 = Off, 1 = Reduced, 2 = Standard, 3 = Increased
uint8       m_safetyCarExperience;      // 0 = Broadcast, 1 = Immersive
uint8       m_formationLap;             // 0 = Off, 1 = On
uint8       m_formationLapExperience;    // 0 = Broadcast, 1 = Immersive
uint8       m_redFlags;                 // 0 = Off, 1 = Reduced, 2 = Standard, 3 = Increased
uint8       m_affectsLicenceLevelSolo;  // 0 = Off, 1 = On
uint8       m_affectsLicenceLevelMP;    // 0 = Off, 1 = On
uint8       m_numSessionsInWeekend;     // Number of session in following array
uint8       m_weekendStructure[12];     // List of session types to show weekend
                                                // structure - see appendix for types

float       m_sector2LapDistanceStart;  // Distance in m around track where sector 2 starts
float       m_sector3LapDistanceStart;  // Distance in m around track where sector 3 starts

// Aero and DRS zones
uint8       m_activeAeroTrackStatus;    // 0 = Full, 1 = Partial
uint8       m_numActiveAeroZonesFull;   // Number of Active Aero zones to follow
ActiveAeroZone m_activeAeroZonesFull[8]; // List of Active Aero zones - max 8
uint8       m_numActiveAeroZonesPartial; // Number of Active Aero zones to follow
ActiveAeroZone m_activeAeroZonesPartial[8]; // List of Active Aero zones - max 8
uint8       m_numDRSZones;              // Number of DRS zones to follow
DRSZone     m_drsZones[4];              // List of DRS zones - max 4
float       m_startReactionTime;        // Driver start reaction time in seconds
                                                // 0.0f if assisted starts

uint8       m_antiLockBrakesAssist;     // 0 = Off, 1 = On
uint8       m_tractionControlAssist;    // 0 = Off, 1 = Medium, 2 = Full
uint8       m_dynamicRacingLineHiVis;   // 0 = Off, 1 = On
uint8       m_dynamicRacingLineColourBlind; // 0 = Off, 1 = Protanopia, 2 = Deuteranopia
                                                // 3 = Tritanopia

uint8       m_recurringRewindPrompt;    // 0 = Off, 1 = On

};
```



Lap Data Packet

The lap data packet gives details of all the cars in the session.

Frequency: Rate as specified in menus

Size: 1399 bytes

Version: 1

```
struct LapData
{
    uint32    m_lastLapTimeInMS;           // Last lap time in milliseconds
    uint32    m_currentLapTimeInMS;        // Current time around the lap in milliseconds
    uint16    m_sector1TimeMSPart;         // Sector 1 time milliseconds part
    uint8     m_sector1TimeMinutesPart;    // Sector 1 whole minute part
    uint16    m_sector2TimeMSPart;         // Sector 2 time milliseconds part
    uint8     m_sector2TimeMinutesPart;    // Sector 2 whole minute part
    uint16    m_deltaToCarInFrontMSPart;   // Time delta to car in front milliseconds part
    uint8     m_deltaToCarInFrontMinutesPart; // Time delta to car in front whole minute part
    uint16    m_deltaToRaceLeaderMSPart;   // Time delta to race leader milliseconds part
    uint8     m_deltaToRaceLeaderMinutesPart; // Time delta to race leader whole minute part
    float     m_lapDistance;               // Distance vehicle is around current lap in metres - could
                                           // be negative if line hasn't been crossed yet
    float     m_totalDistance;             // Total distance travelled in session in metres - could
                                           // be negative if line hasn't been crossed yet
    float     m_safetyCarDelta;            // Delta in seconds for safety car
    uint8     m_carPosition;               // Car race position
    uint8     m_currentLapNum;             // Current lap number
    uint8     m_pitStatus;                 // 0 = none, 1 = pitting, 2 = in pit area
    uint8     m_numPitStops;               // Number of pit stops taken in this race
    uint8     m_sector;                   // 0 = sector1, 1 = sector2, 2 = sector3
    uint8     m_currentLapInvalid;         // Current lap invalid - 0 = valid, 1 = invalid
    uint8     m_penalties;                // Accumulated time penalties in seconds to be added
    uint8     m_totalWarnings;             // Accumulated number of warnings issued
    uint8     m_cornerCuttingWarnings;     // Accumulated number of corner cutting warnings issued
    uint8     m_numUnservicedDriveThroughPens; // Num drive through pens left to serve
    uint8     m_numUnservicedStopGoPens;   // Num stop go pens left to serve
    uint8     m_gridPosition;              // Grid position the vehicle started the race in
    uint8     m_driverStatus;              // Status of driver - 0 = in garage, 1 = flying lap
                                           // 2 = in lap, 3 = out lap, 4 = on track
    uint8     m_resultStatus;              // Result status - 0 = invalid, 1 = inactive, 2 = active
                                           // 3 = finished, 4 = didnotfinish, 5 = disqualified
                                           // 6 = not classified, 7 = retired
    uint8     m_pitLaneTimerActive;        // Pit lane timing, 0 = inactive, 1 = active
    uint16    m_pitLaneTimeInLaneInMS;     // If active, the current time spent in the pit lane in ms
    uint16    m_pitStopTimerInMS;         // Time of the actual pit stop in ms
    uint8     m_pitStopShouldServePen;    // Whether the car should serve a penalty at this stop
    float     m_speedTrapFastestSpeed;     // Fastest speed through speed trap for this car in kmph
    uint8     m_speedTrapFastestLap;       // Lap no the fastest speed was achieved, 255 = not set
};

struct PacketLapData
{
    PacketHeader    m_header;               // Header

    LapData         m_lapData[24];         // Lap data for all cars on track

    uint8           m_timeTrialPBCarIdx;    // Index of Personal Best car in time trial (255 if invalid)
    uint8           m_timeTrialRivalCarIdx; // Index of Rival car in time trial (255 if invalid)
};
```



Event Packet

This packet gives details of events that happen during the course of a session.

Frequency: When the event occurs

Size: 45 bytes

Version: 1

```
// The event details packet is different for each type of event.
// Make sure only the correct type is interpreted.
union EventDataDetails
{
    struct
    {
        uint8  vehicleIdx; // Vehicle index of car achieving fastest lap
        float  lapTime;    // Lap time is in seconds
    } FastestLap;

    struct
    {
        uint8  vehicleIdx; // Vehicle index of car retiring
        uint8  reason;      // Reason - 0 = invalid, 1 = retired, 2 = finished
                                // 3 = terminal damage, 4 = inactive, 5 = not enough laps completed
                                // 6 = black flagged, 7 = red flagged, 8 = mechanical failure
                                // 9 = session skipped, 10 = session simulated
    } Retirement;

    struct
    {
        uint8  reason;      // 0 = Wet track, 1 = Safety car deployed, 2 = Red flag
                                // 3 = Min lap not reached
    } DRSDisabled;

    struct
    {
        uint8  vehicleIdx; // Vehicle index of team mate
    } TeamMateInPits;

    struct
    {
        uint8  vehicleIdx; // Vehicle index of the race winner
    } RaceWinner;

    struct
    {
        uint8  penaltyType;          // Penalty type - see Appendices
        uint8  infringementType;     // Infringement type - see Appendices
        uint8  vehicleIdx;           // Vehicle index of the car the penalty is applied to
        uint8  otherVehicleIdx;      // Vehicle index of the other car involved
        uint8  time;                 // Time gained, or time spent doing action in seconds
        uint8  lapNum;               // Lap the penalty occurred on
        uint8  placesGained;         // Number of places gained by this
    } Penalty;

    struct
    {
        uint8  vehicleIdx;           // Vehicle index of the vehicle triggering speed trap
        float  speed;                // Top speed achieved in kilometres per hour
        uint8  isOverallFastestInSession; // Overall fastest speed in session = 1, otherwise 0
        uint8  isDriverFastestInSession;  // Fastest speed for driver in session = 1, otherwise 0
        uint8  fastestVehicleIdxInSession; // Vehicle index of the vehicle that is the fastest
                                                // in this session
        float  fastestSpeedInSession;      // Speed of the vehicle that is the fastest
                                                // in this session
    } SpeedTrap;

    struct
    {
        uint8  numLights;            // Number of lights showing
    } StartLIghts;
}
```



```
struct
{
    uint8 vehicleIdx;                // Vehicle index of the vehicle serving drive through
} DriveThroughPenaltyServed;

struct
{
    uint8 vehicleIdx;                // Vehicle index of the vehicle serving stop go
    float stopTime;                  // Time spent serving stop go in seconds
} StopGoPenaltyServed;

struct
{
    uint32 flashbackFrameIdentifier; // Frame identifier flashed back to
    float flashbackSessionTime;      // Session time flashed back to
} Flashback;

struct
{
    uint32 buttonStatus;              // Bit flags specifying which buttons are being pressed
                                        // currently - see appendices
} Buttons;

struct
{
    uint8 overtakingVehicleIdx;       // Vehicle index of the vehicle overtaking
    uint8 beingOvertakenVehicleIdx;   // Vehicle index of the vehicle being overtaken
} Overtake;

struct
{
    uint8 safetyCarType;              // 0 = No Safety Car, 1 = Full Safety Car
                                        // 2 = Virtual Safety Car, 3 = Formation Lap Safety Car
    uint8 eventType;                 // 0 = Deployed, 1 = Returning, 2 = Returned
                                        // 3 = Resume Race
} SafetyCar;

struct
{
    uint8 vehicle1Idx;                // Vehicle index of the first vehicle involved in the collision
    uint8 vehicle2Idx;                // Vehicle index of the second vehicle involved in the collision
    uint8 severity;                   // Severity of the collision - 0 = low, 1 = medium, 2 = high
} Collision;

struct
{
    uint8 reason;                     // 0 = Wet track, 1 = Safety car deployed, 2 = Red flag
} PartialModeEnabled;
};

struct PacketEventData
{
    PacketHeader    m_header;          // Header

    uint8           m_eventStringCode[4]; // Event string code, see below
    EventDataDetails m_eventDetails;     // Event details - should be interpreted differently
                                        // for each type
};
```

Event String Codes

Event	Code	Description
Session Started	"SSTA"	Sent when the session starts
Session Ended	"SEND"	Sent when the session ends
Fastest Lap	"FTLP"	When a driver achieves the fastest lap
Retirement	"RTMT"	When a driver retires
DRS enabled	"DRSE"	Race control have enabled DRS



DRS disabled	"DRSD"	Race control have disabled DRS
Team mate in pits	"TMPT"	Your team mate has entered the pits
Chequered flag	"CHQF"	The chequered flag has been waved
Race Winner	"RCWN"	The race winner is announced
Penalty Issued	"PENA"	A penalty has been issued – details in event
Speed Trap Triggered	"SPTP"	Speed trap has been triggered by fastest speed
Start lights	"STLG"	Start lights – number shown
Lights out	"LGOT"	Lights out
Drive through served	"DTSV"	Drive through penalty served
Stop go served	"SGSV"	Stop go penalty served
Flashback	"FLBK"	Flashback activated
Button status	"BUTN"	Button status changed
Red Flag	"RDFL"	Red flag shown
Overtake	"OVTK"	Overtake occurred
Safety Car	"SCAR"	Safety car event – details in event
Collision	"COLL"	Collision between two vehicles has occurred
Partial Mode enabled	"PMEN"	Race control have enabled Partial mode
Partial Mode disabled	"PMDI"	Race control have disabled Partial mode
Overtake enabled	"OVEN"	Race control have enabled Overtake mode
Overtake disabled	"OVDI"	Race control have disabled Overtake mode

Participants Packet

This is a list of participants in the race. If the vehicle is controlled by AI, then the name will be the driver name. If this is a multiplayer game, the names will be the Steam Id on PC, or the LAN name if appropriate.

N.B. on Xbox, the names will always be the driver name, on PlayStation the name will be the LAN name if playing a LAN game, otherwise it will be the driver name.

The array should be indexed by vehicle index.

Frequency: Every 5 seconds

Size: 1470 bytes

Version: 1

```
// RGB value of a colour
struct LiveryColour
{
    uint8    red;
    uint8    green;
    uint8    blue;
};

struct ParticipantData
{
    uint8     m_aiControlled;        // Whether the vehicle is AI (1) or Human (0) controlled
    uint16    m_driverId;            // Driver id - see appendix, 65535 if network human
    uint16    m_networkId;           // Network id - unique identifier for network players
    uint16    m_teamId;              // Team id - see appendix
    uint8     m_myTeam;              // My team flag - 1 = My Team, 0 = otherwise
    uint8     m_raceNumber;          // Race number of the car
    uint8     m_nationality;         // Nationality of the driver
    char       m_name[32];           // Name of participant in UTF-8 format - null terminated
                                           // Will be truncated with ... (U+2026) if too long
    uint8     m_yourTelemetry;       // The player's UDP setting, 0 = restricted, 1 = public
    uint8     m_showOnlineNames;     // The player's show online names setting, 0 = off, 1 = on
    uint16    m_techLevel;           // F1 World tech level
    uint8     m_platform;            // 1 = Steam, 3 = PlayStation, 4 = Xbox, 6 = Origin, 255 = unknown
}
```



```
uint8      m_numColours;          // Number of colours valid for this car
LiveryColour m_liveryColours[4]; // Colours for the car

struct PacketParticipantsData
{
    PacketHeader    m_header;          // Header

    uint8          m_numActiveCars;    // Number of active cars in the data - should match number of
                                        // cars on HUD

    ParticipantData m_participants[24];
};
```

Car Setups Packet

This packet details the car setups for each vehicle in the session. Note that in multiplayer games, other player cars will appear as blank, you will only be able to see your own car setup, regardless of the “Your Telemetry” setting. Spectators will also not be able to see any car setups.

Frequency: 2 per second

Size: 1233 bytes

Version: 1

```
struct CarSetupData
{
    uint8      m_frontWing;          // Front wing aero
    uint8      m_rearWing;           // Rear wing aero
    uint8      m_onThrottle;         // Differential adjustment on throttle (percentage)
    uint8      m_offThrottle;        // Differential adjustment off throttle (percentage)
    float      m_frontCamber;        // Front camber angle (suspension geometry)
    float      m_rearCamber;         // Rear camber angle (suspension geometry)
    float      m_frontToe;           // Front toe angle (suspension geometry)
    float      m_rearToe;            // Rear toe angle (suspension geometry)
    uint8      m_frontSuspension;     // Front suspension
    uint8      m_rearSuspension;     // Rear suspension
    uint8      m_frontAntiRollBar;    // Front anti-roll bar
    uint8      m_rearAntiRollBar;    // Rear anti-roll bar
    uint8      m_frontSuspensionHeight; // Front ride height
    uint8      m_rearSuspensionHeight; // Rear ride height
    uint8      m_brakePressure;       // Brake pressure (percentage)
    uint8      m_brakeBias;           // Brake bias (percentage)
    uint8      m_engineBraking;       // Engine braking (percentage)
    float      m_rearLeftTyrePressure; // Rear left tyre pressure (PSI)
    float      m_rearRightTyrePressure; // Rear right tyre pressure (PSI)
    float      m_frontLeftTyrePressure; // Front left tyre pressure (PSI)
    float      m_frontRightTyrePressure; // Front right tyre pressure (PSI)
    uint8      m_ballast;             // Ballast
    float      m_fuelLoad;            // Fuel load
};

struct PacketCarSetupData
{
    PacketHeader    m_header;          // Header

    CarSetupData    m_carSetups[24];

    float          m_nextFrontWingValue; // Value of front wing after next pit stop - player only
};
```

Car Telemetry Packet



This packet details telemetry for all the cars in the race. It details various values that would be recorded on the car such as speed, throttle application, DRS etc. Note that the rev light configurations are presented separately as well and will mimic real life driver preferences.

Frequency: Rate as specified in menus

Size: 1448 bytes

Version: 1

```
struct CarTelemetryData
{
    uint16    m_speed;                // Speed of car in kilometres per hour
    float     m_throttle;             // Amount of throttle applied (0.0 to 1.0)
    float     m_steer;               // Steering (-1.0 (full lock left) to 1.0 (full lock right))
    float     m_brake;               // Amount of brake applied (0.0 to 1.0)
    uint8     m_clutch;              // Amount of clutch applied (0 to 100)
    int8      m_gear;                // Gear selected (1-8, N=0, R=-1)
    uint16    m_engineRPM;           // Engine RPM
    uint8     m_drs;                 // 0 = off, 1 = on
    uint8     m_revLightsPercent;    // Rev lights indicator (percentage)
    uint16    m_revLightsBitValue;   // Rev lights (bit 0 = leftmost LED, bit 14 = rightmost LED)
    uint16    m_brakesTemperature[4]; // Brakes temperature (celsius)
    uint8     m_tyresSurfaceTemperature[4]; // Tyres surface temperature (celsius)
    uint8     m_tyresInnerTemperature[4]; // Tyres inner temperature (celsius)
    uint8     m_engineTemperature;   // Engine temperature (celsius)
    float     m_tyresPressure[4];     // Tyres pressure (PSI)
    uint8     m_surfaceType[4];       // Driving surface, see appendices
};

struct PacketCarTelemetryData
{
    PacketHeader    m_header;        // Header

    CarTelemetryData m_carTelemetryData[24];

    uint8          m_mfdPanelIndex;  // Index of MFD panel open - 255 = MFD closed
                                        // Single player, race - 0 = Car setup, 1 = Pits
                                        // 2 = Damage, 3 = Engine, 4 = Temperatures
                                        // May vary depending on game mode

    uint8          m_mfdPanelIndexSecondaryPlayer; // See above
    int8           m_suggestedGear;    // Suggested gear for the player (1-8)
                                        // 0 if no gear suggested
};
```

Car Status Packet

This packet details car statuses for all the cars in the race.

Frequency: Rate as specified in menus

Size: 1445 bytes

Version: 1

```
struct CarStatusData
{
    uint8     m_tractionControl;      // Traction control - 0 = off, 1 = medium, 2 = full
    uint8     m_antiLockBrakes;      // 0 (off) - 1 (on)
    uint8     m_fuelMix;              // Fuel mix - 0 = lean, 1 = standard, 2 = rich, 3 = max
    uint8     m_frontBrakeBias;       // Front brake bias (percentage)
    uint8     m_pitLimiterStatus;     // Pit limiter status - 0 = off, 1 = on
    float     m_fuelInTank;           // Current fuel mass
    float     m_fuelCapacity;         // Fuel capacity
    float     m_fuelRemainingLaps;    // Fuel remaining in terms of laps (value on MFD)
    uint16    m_maxRPM;               // Cars max RPM, point of rev limiter
};
```



```
uint16    m_idleRPM;           // Cars idle RPM
uint8     m_maxGears;          // Maximum number of gears
uint8     m_drsAllowed;        // 0 = not allowed, 1 = allowed
uint16    m_drsActivationDistance; // 0 = DRS not available, non-zero - DRS will be available
                                     // in [X] metres
uint8     m_actualTyreCompound; // F1 Modern - 16 = C5, 17 = C4, 18 = C3, 19 = C2, 20 = C1
                                     // 21 = C0, 22 = C6, 7 = inter, 8 = wet
                                     // F1 Classic - 9 = dry, 10 = wet
                                     // F2 - 11 = super soft, 12 = soft, 13 = medium, 14 = hard
                                     // 15 = wet
uint8     m_visualTyreCompound; // F1 visual (can be different from actual compound)
                                     // 16 = soft, 17 = medium, 18 = hard, 7 = inter, 8 = wet
                                     // F1 Classic - same as above
                                     // F2 '20, 15 = wet, 19 = super soft, 20 = soft
                                     // 21 = medium, 22 = hard
uint8     m_tyresAgeLaps;       // Age in laps of the current set of tyres
int8      m_vehicleFiaFlags;    // -1 = invalid/unknown, 0 = none, 1 = green
                                     // 2 = blue, 3 = yellow
float     m_enginePowerICE;      // Engine power output of ICE (W)
float     m_enginePowerMGUK;     // Engine power output of MGU-K (W)
float     m_ersStoreEnergy;      // ERS energy store in Joules
uint8     m_ersDeployMode;      // ERS deployment mode, 0 = none, 1 = medium
                                     // 2 = hotlap, 3 = boost
float     m_ersHarvestedThisLapMGUK; // ERS energy harvested this lap by MGU-K
float     m_ersHarvestedThisLapMGUH; // ERS energy harvested this lap by MGU-H
float     m_ersHarvestedLimitPerLap; // ERS energy harvest limit for this lap
float     m_ersDeployedThisLap;  // ERS energy deployed this lap
uint8     m_networkPaused;      // Whether the car is paused in a network game
};

struct PacketCarStatusData
{
    PacketHeader    m_header;      // Header

    CarStatusData   m_carStatusData[24];
};
```

Final Classification Packet

This packet details the final classification at the end of the race, and the data will match with the post race results screen. This is especially useful for multiplayer games where it is not always possible to send lap times on the final frame because of network delay.

Frequency: Once at the end of a race

Size: 1134 bytes

Version: 1

```
struct FinalClassificationData
{
    uint8     m_position;         // Finishing position
    uint8     m_numLaps;          // Number of laps completed
    uint8     m_gridPosition;     // Grid position of the car
    uint8     m_points;          // Number of points scored
    uint8     m_numPitStops;      // Number of pit stops made
    uint8     m_resultStatus;     // Result status - 0 = invalid, 1 = inactive, 2 = active
                                     // 3 = finished, 4 = didntfinish, 5 = disqualified
                                     // 6 = not classified, 7 = retired
    uint8     m_resultReason;     // Result reason - 0 = invalid, 1 = retired, 2 = finished
                                     // 3 = terminal damage, 4 = inactive, 5 = not enough laps completed
                                     // 6 = black flagged, 7 = red flagged, 8 = mechanical failure
                                     // 9 = session skipped, 10 = session simulated
    uint32    m_bestLapTimeInMS;  // Best lap time of the session in milliseconds
    double    m_totalRaceTime;    // Total race time in seconds without penalties
    uint8     m_penaltiesTime;    // Total penalties accumulated in seconds
    uint8     m_numPenalties;     // Number of penalties applied to this driver
    uint8     m_numTyreStints;    // Number of tyres stints up to maximum
};
```



```
uint8    m_tyreStintsActual[8];    // Actual tyres used by this driver
uint8    m_tyreStintsVisual[8];    // Visual tyres used by this driver
uint8    m_tyreStintsEndLaps[8];    // The lap number stints end on
};

struct PacketFinalClassificationData
{
    PacketHeader    m_header;                // Header

    uint8            m_numCars;                // Number of cars in the final classification
    FinalClassificationData    m_classificationData[24];
};
```

Lobby Info Packet

This packet details the players currently in a multiplayer lobby. It details each player's selected car, any AI involved in the game and also the ready status of each of the participants.

Frequency: Two every second when in the lobby

Size: 1062 bytes

Version: 1

```
struct LobbyInfoData
{
    uint8    m_aiControlled;                // Whether the vehicle is AI (1) or Human (0) controlled
    uint16    m_teamId;                    // Team id - see appendix (65535 if no team currently selected)
    uint8    m_nationality;                // Nationality of the driver
    uint8    m_platform;                    // 1 = Steam, 3 = PlayStation, 4 = Xbox, 6 = Origin, 255 = unknown
    char    m_name[32];                    // Name of participant in UTF-8 format - null terminated
                                                // Will be truncated with ... (U+2026) if too long

    uint8    m_carNumber;                    // Car number of the player
    uint8    m_yourTelemetry;                // The player's UDP setting, 0 = restricted, 1 = public
    uint8    m_showOnlineNames;                // The player's show online names setting, 0 = off, 1 = on
    uint16    m_techLevel;                    // F1 World tech level
    uint8    m_readyStatus;                // 0 = not ready, 1 = ready, 2 = spectating
};

struct PacketLobbyInfoData
{
    PacketHeader    m_header;                // Header

    // Packet specific data
    uint8            m_numPlayers;                // Number of players in the lobby data
    LobbyInfoData    m_lobbyPlayers[24];
};
```

Car Damage Packet

This packet details car damage parameters for all the cars in the race.

Frequency: 10 per second

Size: 1133 bytes

Version: 1

```
struct CarDamageData
{
    float    m_tyresWear[4];                // Tyre wear (percentage)
    uint8    m_tyresDamage[4];                // Tyre damage (percentage)
    uint8    m_brakesDamage[4];                // Brakes damage (percentage)
    uint8    m_tyreBlisters[4];                // Tyre blisters value (percentage)
    uint8    m_frontLeftWingDamage;                // Front left wing damage (percentage)
    uint8    m_frontRightWingDamage;                // Front right wing damage (percentage)
    uint8    m_rearWingDamage;                // Rear wing damage (percentage)
};
```



```
uint8    m_floorDamage;           // Floor damage (percentage)
uint8    m_diffuserDamage;        // Diffuser damage (percentage)
uint8    m_sidepodDamage;         // Sidepod damage (percentage)
uint8    m_drsFault;              // Indicator for DRS fault, 0 = OK, 1 = fault
uint8    m_ersFault;              // Indicator for ERS fault, 0 = OK, 1 = fault
uint8    m_gearBoxDamage;         // Gear box damage (percentage)
uint8    m_engineDamage;          // Engine damage (percentage)
uint8    m_engineMGUHWear;        // Engine wear MGU-H (percentage)
uint8    m_engineESWear;          // Engine wear ES (percentage)
uint8    m_engineCEWear;          // Engine wear CE (percentage)
uint8    m_engineICEWear;         // Engine wear ICE (percentage)
uint8    m_engineMGUKWear;        // Engine wear MGU-K (percentage)
uint8    m_engineTCWear;          // Engine wear TC (percentage)
uint8    m_engineBlown;           // Engine blown, 0 = OK, 1 = fault
uint8    m_engineSeized;          // Engine seized, 0 = OK, 1 = fault
}

struct PacketCarDamageData
{
    PacketHeader    m_header;           // Header

    CarDamageData   m_carDamageData[24];
};
```

Session History Packet

This packet contains lap times and tyre usage for the session. **This packet works slightly differently to other packets. To reduce CPU and bandwidth, each packet relates to a specific vehicle and is sent every 1/20 s, and the vehicle being sent is cycled through. Therefore in a 20 car race you should receive an update for each vehicle at least once per second.**

Note that at the end of the race, after the final classification packet has been sent, a final bulk update of all the session histories for the vehicles in that session will be sent.

Frequency: 20 per second but cycling through cars

Size: 1460 bytes

Version: 1

```
struct LapHistoryData
{
    uint32    m_lapTimeInMS;           // Lap time in milliseconds
    uint16    m_sector1TimeMSPart;     // Sector 1 milliseconds part
    uint8     m_sector1TimeMinutesPart; // Sector 1 whole minute part
    uint16    m_sector2TimeMSPart;     // Sector 2 time milliseconds part
    uint8     m_sector2TimeMinutesPart; // Sector 2 whole minute part
    uint16    m_sector3TimeMSPart;     // Sector 3 time milliseconds part
    uint8     m_sector3TimeMinutesPart; // Sector 3 whole minute part
    uint8     m_lapValidBitFlags;       // 0x01 bit set-lap valid, 0x02 bit set-sector 1 valid
                                           // 0x04 bit set-sector 2 valid, 0x08 bit set-sector 3 valid
};

struct TyreStintHistoryData
{
    uint8     m_endLap;                 // Lap the tyre usage ends on (255 of current tyre)
    uint8     m_tyreActualCompound;     // Actual tyres used by this driver
    uint8     m_tyreVisualCompound;     // Visual tyres used by this driver
};

struct PacketSessionHistoryData
{
    PacketHeader    m_header;           // Header

    uint8          m_carIdx;             // Index of the car this lap data relates to
    uint8          m_numLaps;            // Num laps in the data (including current partial lap)
    uint8          m_numTyreStints;      // Number of tyre stints in the data
};
```



```
uint8      m_bestLapTimeLapNum;      // Lap the best lap time was achieved on
uint8      m_bestSector1LapNum;      // Lap the best Sector 1 time was achieved on
uint8      m_bestSector2LapNum;      // Lap the best Sector 2 time was achieved on
uint8      m_bestSector3LapNum;      // Lap the best Sector 3 time was achieved on

LapHistoryData  m_lapHistoryData[100]; // 100 laps of data max
TyreStintHistoryData m_tyreStintsHistoryData[8];

};
```

Tyre Sets Packet

This packets gives a more in-depth details about tyre sets assigned to a vehicle during the session.

Frequency: 20 per second but cycling through cars

Size: 231 bytes

Version: 1

```
struct TyreSetData
{
    uint8      m_actualTyreCompound; // Actual tyre compound used
    uint8      m_visualTyreCompound; // Visual tyre compound used
    uint8      m_wear;               // Tyre wear (percentage)
    uint8      m_available;          // Whether this set is currently available
    uint8      m_recommendedSession; // Recommended session for tyre set, see appendix
    uint8      m_lifeSpan;           // Laps left in this tyre set
    uint8      m_usableLife;         // Max number of laps recommended for this compound
    int16      m_lapDeltaTime;       // Lap delta time in milliseconds compared to fitted set
    uint8      m_fitted;             // Whether the set is fitted or not
};

struct PacketTyreSetsData
{
    PacketHeader  m_header;          // Header

    uint8         m_carIdx;          // Index of the car this data relates to

    TyreSetData   m_tyreSetData[20]; // 13 (dry) + 7 (wet)

    uint8         m_fittedIdx;       // Index into array of fitted tyre
};
```



Motion Ex Packet

The motion packet gives extended data for the car being driven with the goal of being able to drive a motion platform setup.

Frequency: Rate as specified in menus

Size: 273 bytes

Version: 1

```
struct PacketMotionExData
{
    PacketHeader    m_header;                // Header

    // Extra player car ONLY data
    float           m_suspensionPosition[4]; // Note: All wheel arrays have the following order:
    float           m_suspensionVelocity[4];  // RL, RR, FL, FR
    float           m_suspensionAcceleration[4]; // RL, RR, FL, FR
    float           m_wheelSpeed[4];          // Speed of each wheel
    float           m_wheelSlipRatio[4];      // Slip ratio for each wheel
    float           m_wheelSlipAngle[4];      // Slip angles for each wheel
    float           m_wheelLatForce[4];       // Lateral forces for each wheel
    float           m_wheelLongForce[4];      // Longitudinal forces for each wheel
    float           m_heightOfCOGAboveGround; // Height of centre of gravity above ground
    float           m_localVelocityX;         // Velocity in local space - metres/s
    float           m_localVelocityY;         // Velocity in local space
    float           m_localVelocityZ;         // Velocity in local space
    float           m_angularVelocityX;       // Angular velocity x-component - radians/s
    float           m_angularVelocityY;       // Angular velocity y-component
    float           m_angularVelocityZ;       // Angular velocity z-component
    float           m_angularAccelerationX;   // Angular acceleration x-component - radians/s/s
    float           m_angularAccelerationY;   // Angular acceleration y-component
    float           m_angularAccelerationZ;   // Angular acceleration z-component
    float           m_frontWheelsAngle;       // Current front wheels angle in radians
    float           m_wheelVertForce[4];      // Vertical forces for each wheel
    float           m_frontAeroHeight;        // Front plank edge height above road surface
    float           m_rearAeroHeight;        // Rear plank edge height above road surface
    float           m_frontRollAngle;         // Roll angle of the front suspension
    float           m_rearRollAngle;         // Roll angle of the rear suspension
    float           m_chassisYaw;             // Yaw angle of the chassis relative to the direction
                                                // of motion - radians
    float           m_chassisPitch;           // Pitch angle of the chassis relative to the
                                                // direction of motion - radians
    float           m_wheelCamber[4];         // Camber of each wheel in radians
    float           m_wheelCamberGain[4];     // Camber gain for each wheel in radians, difference
                                                // between active camber and dynamic camber
};
```

Time Trial Packet

The time trial data gives extra information only relevant to time trial game mode. This packet will not be sent in other game modes.

Frequency: 1 per second

Size: 104 bytes

Version: 1

```
struct TimeTrialDataSet
{
    uint8          m_carIdx;                // Index of the car this data relates to
    uint16         m_teamId;                // Team id - see appendix
    uint32         m_lapTimeInMS;           // Lap time in milliseconds
    uint32         m_sector1TimeInMS;       // Sector 1 time in milliseconds
    uint32         m_sector2TimeInMS;       // Sector 2 time in milliseconds
};
```




```

uint32    m_sector3TimeInMS;           // Sector 3 time in milliseconds
uint8     m_tractionControl;           // 0 = assist off, 1 = assist on
uint8     m_gearboxAssist;              // 0 = assist off, 1 = assist on
uint8     m_antiLockBrakes;            // 0 = assist off, 1 = assist on
uint8     m_equalCarPerformance;       // 0 = Realistic, 1 = Equal
uint8     m_customSetup;               // 0 = No, 1 = Yes
uint8     m_valid;                     // 0 = invalid, 1 = valid
};

struct PacketTimeTrialData
{
    PacketHeader    m_header;           // Header

    TimeTrialDataSet    m_playerSessionBestDataSet;    // Player session best data set
    TimeTrialDataSet    m_personalBestDataSet;        // Personal best data set
    TimeTrialDataSet    m_rivalDataSet;               // Rival data set
};

```

Lap Positions Packet

The lap positions data indicates which position each car was on at the start of each lap. Using this information a lap positions chart can be constructed. Note that only a maximum of 50 laps will be transmitted in a packet. If more than 50 laps have occurred then two packets will be transmitted, with different m_lapStart parameters. The whole lap position history can be recreated merging both of these.

Frequency: 1 per second

Size: 1231 bytes

Version: 1

```

struct PacketLapPositionsData
{
    PacketHeader    m_header;           // Header

    // Packet specific data
    uint8           m_numLaps;           // Number of laps in the data
    uint8           m_lapStart;          // Index of the lap where the data starts, 0 indexed

    // Array holding the position of the car in a given lap, 0 if no record
    uint8           m_positionForVehicleIdx[50][24];
};

```

Car Telemetry 2 Packet

This packet details additional telemetry for all the cars in the race, as the initial car telemetry was getting too large.

Frequency: Rate as specified in menus

Size: 269 bytes

Version: 1

```

struct CarTelemetry2Data
{
    uint8          m_activeAeroMode;     // 0 = Corner mode, 1 = Straight mode
    uint8          m_activeAeroAvailable; // 0 = not available, 1 = available
    uint16         m_activeAeroActivationDistance; // 0 = Active aero not available, non-zero - Active
                                                    // aero will be available in [X] metres
    uint8          m_overtakeAvailable;  // 0 = not available, 1 = available
    uint8          m_overtakeActive;     // 0 = not active, 1 = active
    uint16         m_overtakeActivationDistance; // 0 = Overtake Mode not available, non-zero -

```



```
uint8      m_2026Regulations;           // Overtake Mode will be available in [X] metres
uint8      m_drivingWrongWay;           // 0 = vehicle conforms to pre-2026, 1 = 2026
};                                         // regulations applicable
                                           // Whether the car is driving the wrong way

struct PacketCarTelemetry2Data
{
    PacketHeader      m_header;           // Header

    CarTelemetry2Data m_carTelemetry2Data[24];
};
```



Restricted data (Your Telemetry setting)

There is some data in the UDP that you may not want other players seeing if you are in a multiplayer game. This is controlled by the "Your Telemetry" setting in the Telemetry options. The options are:

- Restricted (Default) – other players viewing the UDP data will not see values for your car
- Public – all other players can see all the data for your car

Note: You can always see the data for the car you are driving regardless of the setting.

The following data items are set to zero if the player driving the car in question has their "Your Telemetry" set to "Restricted":

Car status packet

- m_fuelInTank
- m_fuelCapacity
- m_fuelMix
- m_fuelRemainingLaps
- m_frontBrakeBias
- m_ersDeployMode
- m_ersStoreEnergy
- m_ersDeployedThisLap
- m_ersHarvestedThisLapMGUK
- m_ersHarvestedThisLapMGUH
- m_enginePowerICE
- m_enginePowerMGUK

Car damage packet

- m_frontLeftWingDamage
- m_frontRightWingDamage
- m_rearWingDamage
- m_floorDamage
- m_diffuserDamage
- m_sidepodDamage
- m_engineDamage
- m_gearBoxDamage
- m_tyresWear (All four wheels)
- m_tyresDamage (All four wheels)
- m_brakesDamage (All four wheels)
- m_drsFault
- m_engineMGUHWear
- m_engineESWear
- m_engineCEWear
- m_engineICEWear
- m_engineMGUKWear



- m_engineTCWear

Tyre set packet

- All data within this packet for player car

To allow other players to view your online ID in their UDP output during an online session, you must enable the "Show online ID / gamertags" option. Selecting this will bring up a confirmation box that must be confirmed before this option is enabled.

Please note that all options can be changed during a game session and will take immediate effect.

FAQS

How do I enable the UDP Telemetry Output?

In F1 26, UDP telemetry output is controlled via the in-game menus. To enable this, enter the options menu from the main menu (triangle / Y), then enter the settings menu - the UDP option will be at the bottom of the list. From there you will be able to enable / disable the UDP output, configure the IP address and port for the receiving application, toggle broadcast mode and set the send rate. Broadcast mode transmits the data across the network subnet to allow multiple devices on the same subnet to be able to receive this information. When using broadcast mode it is not necessary to set a target IP address, just a target port for applications to listen on.

Advanced PC Users: You can additionally edit the game's configuration XML file to configure UDP output. The file is located here (after an initial boot of the game):

```
...\Documents\My Games\<game_folder>\hardwaresettings\hardware_settings_config.xml
```

You should see the tag:

```
<motion>
...
  <udp enabled="false" broadcast="false" ip="127.0.0.1" port="20777" sendRate="20"
format="2026" yourTelemetry="restricted" onlineNames="off" />
...
</motion>
```

Here you can set the values manually. Note that any changes made within the game when it is running will overwrite any changes made manually. Note the enabled flag is now a state.

What has changed since last year?

F1® 25 2026 SEASON DLC sees the following changes to the UDP specification:

- **[IMPORTANT]** Increasing the maximum number of vehicles to 24 to accommodate the new team for the 2026 season, so most packet sizes have changed
- Engine temperature has been changed to a uint8 as it didn't need to be a uint16
- G-force values have been quantised to reduce space in the motion packet as the number of cars are increasing
- Added new tracks, teams, drivers and cars for 2026 season
- Increased the size of Driver, Network and Team Ids to allow for a larger database

- Added new Car Telemetry 2 packet for additional data regarding 2026 regulations and driving the wrong way
- Added activation zones and active aero status to session packet
- Added ERS harvest limit per lap to car status packet
- Added driver reaction time for race starts
- Added collision severity to event
- Added anti-lock brakes, traction control, hi-vis and colourblind assists & rewind prompt options
- Race control messages for partial mode and overtake mode activation have been added to the event packet

What is the order of the wheel arrays?

All wheel arrays are in the following order:

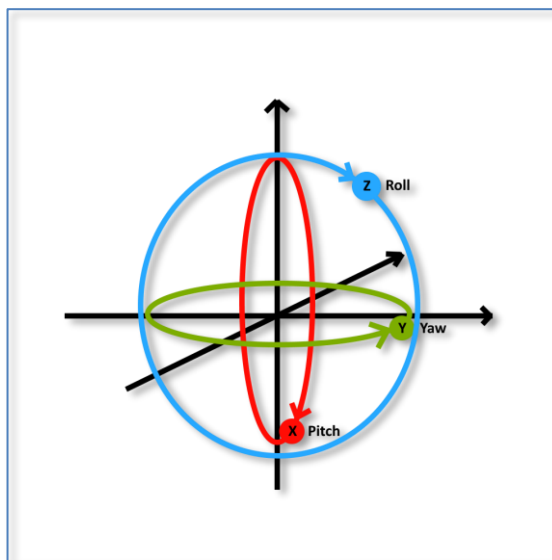
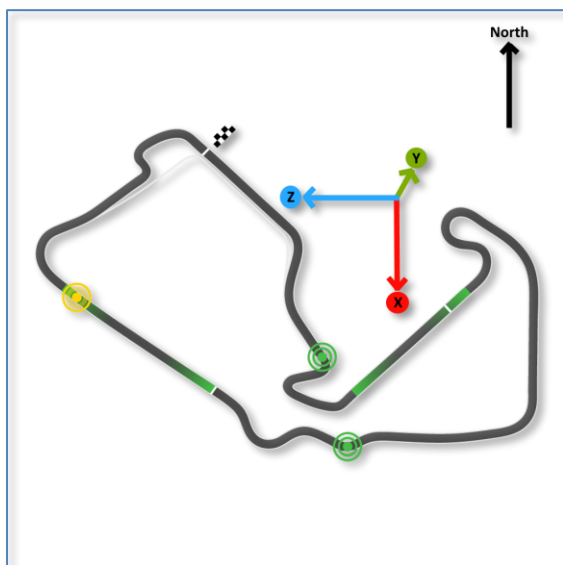
- 0 - Rear Left (RL)
- 1 - Rear Right (RR)
- 2 - Front Left (FL)
- 3 - Front Right (FR)

Do the vehicle indices change?

During a session, each car is assigned a vehicle index. This will not change throughout the session and all the arrays that are sent use this vehicle index to dereference the correct piece of data.

What are the co-ordinate systems used?

Here is a visual representation of the co-ordinate system used with the F1 telemetry data.



What encoding format is used?

All values are encoded using Little Endian format.

Are the data structures packed?

Yes, all data is packed, there is no padding used.



How many cars are in the data structures?

The maximum number of cars in the data structures is 24, to allow for certain game modes, although the data is not always filled in.

You should always check the data item called `m_numActiveCars` in the participants packet which tells you how many cars are active in the race. However, you should check the individual result status of each car in the lap data to see if that car is actively providing data. If it is not `"Invalid"` or `"Inactive"` then the corresponding vehicle index has valid data.

How often are updated packets sent?

For the packets which get updated at "Rate as specified in the menus" you can be guaranteed that on the frame that these get sent they will all get sent together and will never be separated across frames. This of course relies on the reliability of your network as to whether they are received correctly as everything is sent via UDP. Other packets that get sent at specific rates can arrive on any frame.

If you are connected to the game when it starts transmitting the first frame will contain the following information to help initialise data structures on the receiving application:

Packets sent on Frame 1: (All packets sent on this frame have "Session timestamp" 0.000)

- Session
- Participants
- Car Setups
- Lap Data
- Motion Data
- Car Telemetry
- Car Status
- Car Damage
- Motion Ex Data

As an example, assuming that you are running at 60Hz with 60Hz update rate selected in the menus then you would expect to see the following packets and timestamps:

Packets sent on Frame 2: (All packets sent on this frame have "Session timestamp" 0.016)

- Lap Data
- Motion Data
- Car Telemetry
- Car Status
- Motion Ex Data

...

Packets sent on Frame 31: (All packets sent on this frame have "Session timestamp" 0.5)

- Session (since 2 updates per second)



- Car Setups (since 2 updates per second)
- Lap Data
- Motion Data
- Car Telemetry
- Car Status
- Car Damage (since 2 updates per second)
- Motion Ex Data

Will my old app still work with F1 26?

Please note that F1 26 will only support the previous 2 UDP formats.

F1 26 uses a new format for the UDP data. However, some earlier formats of the data are still supported so that most older apps implemented using the previous data formats should work with little or no change from the developer. To use the old formats, please enter the UDP options menu and set "UDP Format" to either "2025" or "2024".

Specifications for the older formats can be seen here:

- F1 24 - <https://answers.ea.com/t5/General-Discussion/F1-24-UDP-Specification/td-p/13745220>
- F1 25 - <https://forums.ea.com/blog/f1-games-game-info-hub-en/ea-sports%E2%84%A2-f1%C2%AE25-udp-specification/12187347>

How do I enable D-BOX output?

D-BOX output is currently supported on the PC platform. In F1 26, the D-BOX activation can be controlled via the menus. Navigate to [Game Options->Settings->UDP Telemetry Settings->D-BOX](#) to activate this on your system.

Advanced PC Users: It is possible to control D-BOX by editing the games' configuration XML file. The file is located here (after an initial boot of the game):

```
...\Documents\My Games\<game_folder>\hardwaresettings\hardware_settings_config.xml
```

You should see the tag:

```
<motion>
  <dbox enabled="false" />
  ...
</motion>
```

Set the "enabled" value to "true" to allow the game to output to your D-BOX motion platform. Note that any changes made within the game when it is running will overwrite any changes made manually.

How can I disable in-game support for LED device?

The F1 game has native support for some of the basic features supported by some external LED devices, such as the *Leo Bodnar SLI Pro* and the *Fanatec* steering wheels. To avoid conflicts between the game's implementation and any third-party device managers on the PC platform it may be necessary to disable the native support. This is done using the following [led_display](#) flags in the [hardware_settings_config.xml](#). The file is located here (after an initial boot of the game):

```
...\Documents\My Games\<game_folder>\hardwaresettings\hardware_settings_config.xml
```



The flags to enable/disable LED output are:

```
<led_display fanatecNativeSupport="true" sliProNativeSupport="true" />
```

The `sliProNativeSupport` flag controls the output to SLI Pro devices. The `fanatecNativeSupport` flag controls the output to Fanatec (and some related) steering wheel LEDs. Set the values for any of these to `"false"` to disable them and avoid conflicts with your own device manager.

Please note there is an additional flag to manually control the LED brightness on the SLI Pro:

```
<led_display sliProForceBrightness="127" />
```

This option (using value in the range 0-255) will be ignored when setting the `sliProNativeSupport` flag to `"false"`.

Also note it is now possible to edit these values on the fly via the [Game Options->Settings->UDP Telemetry Settings](#) menu.

Can I configure the UDP output using an XML File?

PC users can edit the game's configuration XML file to configure UDP output. The file is located here (after an initial boot of the game):

```
...\Documents\My Games\<game_folder>\hardwaresettings\hardware_settings_config.xml
```

You should see the tag:

```
<motion>
...
  <udp enabled="false" broadcast="false" ip="127.0.0.1" port="20777" sendRate="20"
format="2026" yourTelemetry="restricted" onlineNames="off" />
...
</motion>
```

Here you can set the values manually. Note that any changes made within the game when it is running will overwrite any changes made manually.



Appendices

Here are the values used for some of the parameters in the UDP data output.

Team IDs

ID	Team	ID	Team
0	Mercedes	465	Art GP '25
1	Ferrari	466	Campos '25
2	Red Bull Racing	467	Rodin Motorsport '25
3	Williams	468	AIX Racing '25
4	Aston Martin	469	DAMS '25
5	Alpine	470	Hitech '25
6	RB	471	MP Motorsport '25
7	Haas	472	Prema '25
8	McLaren	473	Trident '25
9	Sauber	474	Van Amersfoort Racing '25
41	F1 Generic	475	Invicta '25
104	F1 Custom Team	476	Mercedes '26
129	Konnersport	477	Ferrari '26
142	APXGP '24	478	Red Bull Racing '26
154	APXGP '25	479	Williams '26
155	Konnersport '24	480	Aston Martin '26
158	Art GP '24	481	Alpine '26
159	Campos '24	482	RB '26
160	Rodin Motorsport '24	483	Haas '26
161	AIX Racing '24	484	McLaren '26
162	DAMS '24	485	Audi '26
163	Hitech '24	486	Cadillac '26
164	MP Motorsport '24	489	Art GP '26
165	Prema '24	490	Campos '26
166	Trident '24	491	Rodin Motorsport '26
167	Van Amersfoort Racing '24	492	AIX Racing '26
168	Invicta '24	493	DAMS '26
185	Mercedes '24	494	Hitech '26
186	Ferrari '24	495	MP Motorsport '26
187	Red Bull Racing '24	496	Prema '26
188	Williams '24	497	Trident '26
189	Aston Martin '24	498	Van Amersfoort Racing '26
190	Alpine '24	499	Invicta '26
191	RB '24		
192	Haas '24		
193	McLaren '24		
194	Sauber '24		

Driver IDs

ID	Driver	ID	Driver	ID	Driver
0	Carlos Sainz	62	Alexander Albon	164	Joshua Dürksen
2	Daniel Ricciardo	70	Rashid Nair	165	Andrea-Kimi Antonelli
3	Fernando Alonso	71	Jack Tremblay	166	Ritomo Miyata
4	Felipe Massa	77	Ayrton Senna	167	Rafael Villagómez
7	Lewis Hamilton	80	Guanyu Zhou	168	Zak O'Sullivan
9	Max Verstappen	83	Juan Manuel Correa	169	Pepe Marti
10	Nico Hülkenberg	90	Michael Schumacher	170	Sonny Hayes
11	Kevin Magnussen	94	Yuki Tsunoda	171	Joshua Pearce
14	Sergio Pérez	102	Aidan Jackson	172	Callum Voisin
15	Valtteri Bottas	109	Jenson Button	173	Matias Zagazeta
17	Esteban Ocon	110	David Coulthard	174	Nikola Tsolov
19	Lance Stroll	112	Oscar Piastri	175	Tim Tramnitz
20	Arron Barnes	113	Liam Lawson	185	Luca Cortez
21	Martin Giles	116	Richard Verschoor	186	Luke Browning
22	Alex Murray	123	Enzo Fittipaldi	187	Cian Shields
23	Lucas Roth	125	Mark Webber	188	Arvid Lindblad
24	Igor Correia	126	Jacques Villeneuve	189	Dino Beganovic
25	Sophie Levasseur	127	Callie Mayer	190	Leonardo Fornaroli
26	Jonas Schiffer	132	Logan Sargeant	191	Oliver Goethe
27	Alain Forest	136	Jack Doohan	192	Gabriele Mini
28	Jay Letourneau	137	Amaury Cordeel	193	Sebastián Montoya
29	Esto Saari	138	Dennis Hauger	194	Alexander Dunne
30	Yasar Atiyeh	145	Zane Maloney	195	Max Esterson
31	Callisto Calabresi	146	Victor Martins	196	Sami Meguetounif
32	Naota Izumi	147	Oliver Bearman	197	John Bennett
33	Howard Clarke	148	Jak Crawford	198	Emerson Fittipaldi
34	Lars Kaufmann	149	Isack Hadjar	199	Tasanopol Inthraphuvasak
35	Marie Laursen	152	Roman Stanek	200	Noel León
36	Flavio Nieves	153	Kush Maini	201	Roman Bilinski
38	Klimek Michalski	156	Brendon Leigh	202	Colton Herta
39	Santiago Moreno	157	David Tonizza	203	Rafael Câmara
40	Benjamin Coppens	158	Jarno Opmeer	204	Mari Boya
41	Noah Visser	159	Lucas Blakeley	205	Martinius Stenshorne
50	George Russell	160	Paul Aron	206	Laurens van Hoepen
54	Lando Norris	161	Gabriel Bortoletto	207	Nico Varrone
58	Charles Leclerc	162	Franco Colapinto		
59	Pierre Gasly	163	Taylor Barnard		



Track IDs

ID	Track
0	Melbourne
2	Shanghai
3	Sakhir (Bahrain)
4	Catalunya
5	Monaco
6	Montreal
7	Silverstone
9	Hungaroring
10	Spa
11	Monza
12	Singapore
13	Suzuka
14	Abu Dhabi
15	Texas
16	Brazil
17	Austria
19	Mexico
20	Baku (Azerbaijan)
26	Zandvoort
27	Imola
29	Jeddah
30	Miami
31	Las Vegas
32	Losail
39	Silverstone (Reverse)
40	Austria (Reverse)
41	Zandvoort (Reverse)
42	Madrid

Nationality IDs

ID	Nationality	ID	Nationality	ID	Nationality
1	American	31	Greek	61	Paraguayan
2	Argentinean	32	Guatemalan	62	Peruvian
3	Australian	33	Honduran	63	Polish
4	Austrian	34	Hong Konger	64	Portuguese
5	Azerbaijani	35	Hungarian	65	Qatari
6	Bahraini	36	Icelander	66	Romanian
7	Belgian	37	Indian	68	Salvadoran
8	Bolivian	38	Indonesian	69	Saudi
9	Brazilian	39	Irish	70	Scottish
10	British	40	Israeli	71	Serbian
11	Bulgarian	41	Italian	72	Singaporean
12	Cameroonian	42	Jamaican	73	Slovakian
13	Canadian	43	Japanese	74	Slovenian
14	Chilean	44	Jordanian	75	South Korean
15	Chinese	45	Kuwaiti	76	South African
16	Colombian	46	Latvian	77	Spanish
17	Costa Rican	47	Lebanese	78	Swedish
18	Croatian	48	Lithuanian	79	Swiss
19	Cypriot	49	Luxembourger	80	Thai
20	Czech	50	Malaysian	81	Turkish
21	Danish	51	Maltese	82	Uruguayan
22	Dutch	52	Mexican	83	Ukrainian
23	Ecuadorian	53	Monegasque	84	Venezuelan
24	English	54	New Zealander	85	Barbadian
25	Emirian	55	Nicaraguan	86	Welsh
26	Estonian	56	Northern Irish	87	Vietnamese
27	Finnish	57	Norwegian	88	Algerian
28	French	58	Omani	89	Bosnian
29	German	59	Pakistani	90	Filipino
30	Ghanaian	60	Panamanian		



Game Mode IDs

ID	Mode
4	Grand Prix '23
5	Time Trial
6	Splitscreen
7	Online Custom
15	Online Weekly Event
17	Story Mode (Braking Point)
27	My Team Career '25
28	Driver Career '25
29	Career '25 Online
30	Challenge Career '25
75	Story Mode (APXGP)
127	Benchmark

Session types

ID	Session type
0	Unknown
1	Practice 1
2	Practice 2
3	Practice 3
4	Short Practice
5	Qualifying 1
6	Qualifying 2
7	Qualifying 3
8	Short Qualifying
9	One-Shot Qualifying
10	Sprint Shootout 1
11	Sprint Shootout 2
12	Sprint Shootout 3
13	Short Sprint Shootout
14	One-Shot Sprint Shootout
15	Race
16	Race 2
17	Race 3
18	Time Trial

Ruleset IDs

ID	Ruleset
0	Practice & Qualifying
1	Race



2	Time Trial
12	Elimination

Surface types

These types are from physics data and show what type of contact each wheel is experiencing.

ID	Surface
0	Tarmac
1	Rumble strip
2	Concrete
3	Rock
4	Gravel
5	Mud
6	Sand
7	Grass
8	Water
9	Cobblestone
10	Metal
11	Ridged

Button flags

These flags are used in the telemetry packet to determine if any buttons are being held on the controlling device. If the value below logical ANDed with the button status is set then the corresponding button is being held.

Bit Flag	Button
0x00000001	Cross or A
0x00000002	Triangle or Y
0x00000004	Circle or B
0x00000008	Square or X
0x00000010	D-pad Left
0x00000020	D-pad Right
0x00000040	D-pad Up
0x00000080	D-pad Down
0x00000100	Options or Menu
0x00000200	L1 or LB
0x00000400	R1 or RB
0x00000800	L2 or LT
0x00001000	R2 or RT
0x00002000	Left Stick Click
0x00004000	Right Stick Click
0x00008000	Right Stick Left
0x00010000	Right Stick Right



0x00020000	Right Stick Up
0x00040000	Right Stick Down
0x00080000	Special
0x00100000	UDP Action 1
0x00200000	UDP Action 2
0x00400000	UDP Action 3
0x00800000	UDP Action 4
0x01000000	UDP Action 5
0x02000000	UDP Action 6
0x04000000	UDP Action 7
0x08000000	UDP Action 8
0x10000000	UDP Action 9
0x20000000	UDP Action 10
0x40000000	UDP Action 11
0x80000000	UDP Action 12

Penalty types

ID	Penalty meaning
0	Drive through
1	Stop Go
2	Grid penalty
3	Penalty reminder
4	Time penalty
5	Warning
6	Disqualified
7	Removed from formation lap
8	Parked too long timer
9	Tyre regulations
10	This lap invalidated
11	This and next lap invalidated
12	This lap invalidated without reason
13	This and next lap invalidated without reason
14	This and previous lap invalidated
15	This and previous lap invalidated without reason
16	Retired
17	Black flag timer

Infringement types

ID	Infringement meaning
0	Blocking by slow driving
1	Blocking by wrong way driving



2	Reversing off the start line
3	Big Collision
4	Small Collision
5	Collision failed to hand back position single
6	Collision failed to hand back position multiple
7	Corner cutting gained time
8	Corner cutting overtake single
9	Corner cutting overtake multiple
10	Crossed pit exit lane
11	Ignoring blue flags
12	Ignoring yellow flags
13	Ignoring drive through
14	Too many drive throughs
15	Drive through reminder serve within n laps
16	Drive through reminder serve this lap
17	Pit lane speeding
18	Parked for too long
19	Ignoring tyre regulations
20	Too many penalties
21	Multiple warnings
22	Approaching disqualification
23	Tyre regulations select single
24	Tyre regulations select multiple
25	Lap invalidated corner cutting
26	Lap invalidated running wide
27	Corner cutting ran wide gained time minor
28	Corner cutting ran wide gained time significant
29	Corner cutting ran wide gained time extreme
30	Lap invalidated wall riding
31	Lap invalidated flashback used
32	Lap invalidated reset to track
33	Blocking the pitlane
34	Jump start
35	Safety car to car collision
36	Safety car illegal overtake
37	Safety car exceeding allowed pace
38	Virtual safety car exceeding allowed pace
39	Formation lap below allowed speed
40	Formation lap parking
41	Retired mechanical failure
42	Retired terminally damaged
43	Safety car falling too far back
44	Black flag timer
45	Unserved stop go penalty
46	Unserved drive through penalty



47	Engine component change
48	Gearbox change
49	Parc Fermé change
50	League grid penalty
51	Retry penalty
52	Illegal time gain
53	Mandatory pitstop
54	Attribute assigned

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